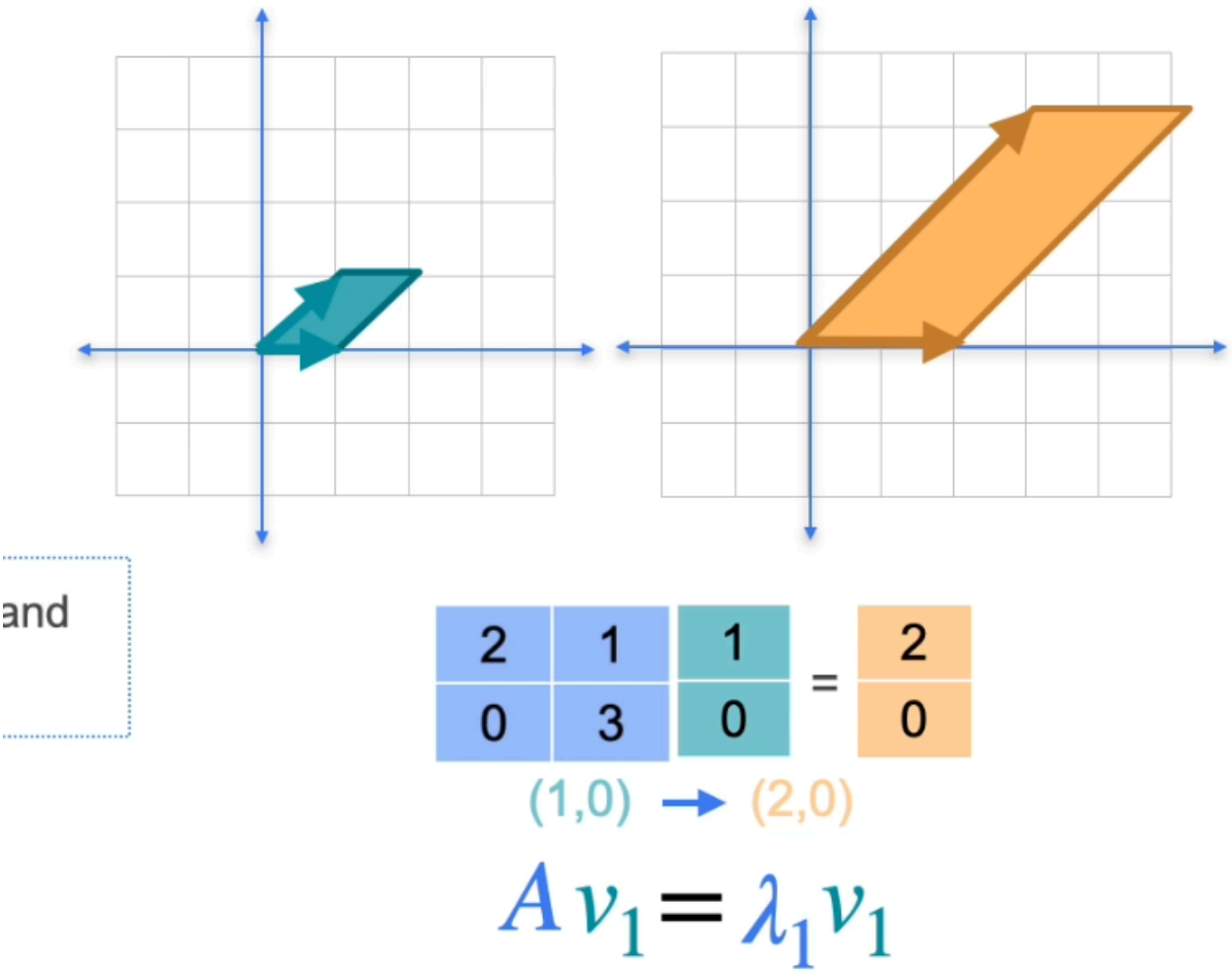
Eigenvalues and Eigenvectors

An eigenvector is the direction that a matrix points to.

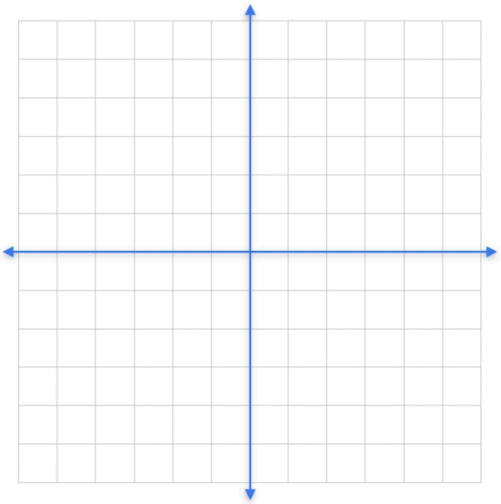


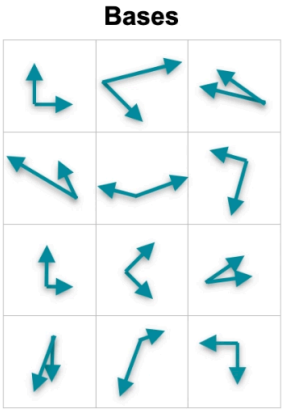
Bases (Plural form of Basis)

A basis is a set of linearly independent vectors that spans a complete vector space, acting as fundamental building blocks to uniquely represent any vectors within the space through linear combinations.

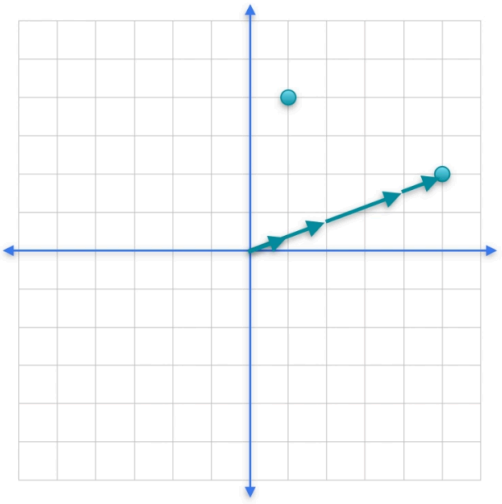
- All bases for a space have the same number of vectors, known as the dimension.

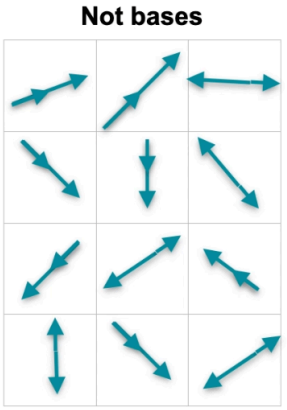
Bases (plural of “basis”)





What is not a basis?

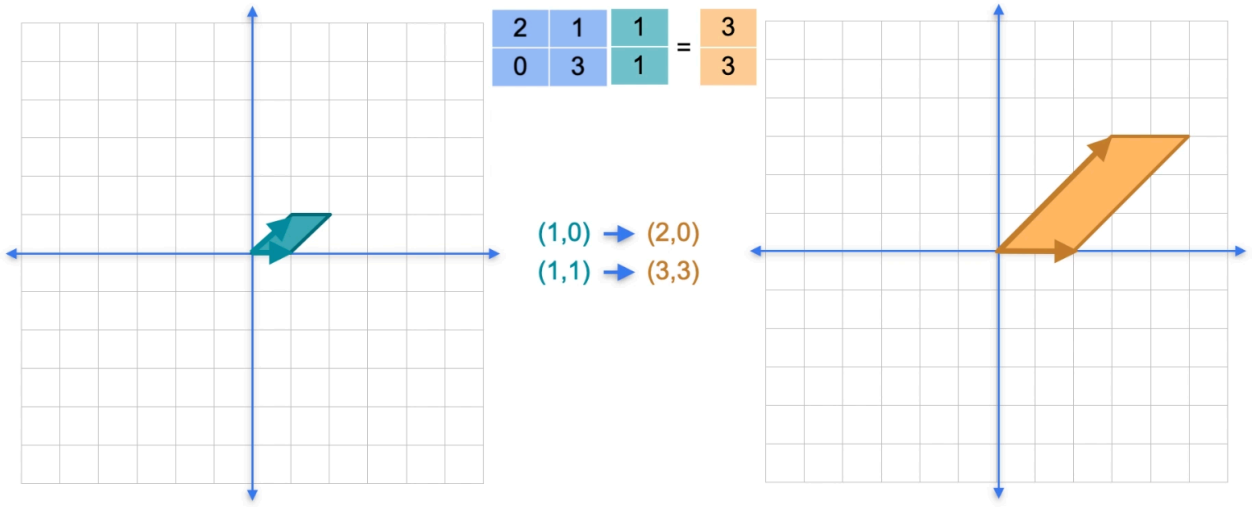




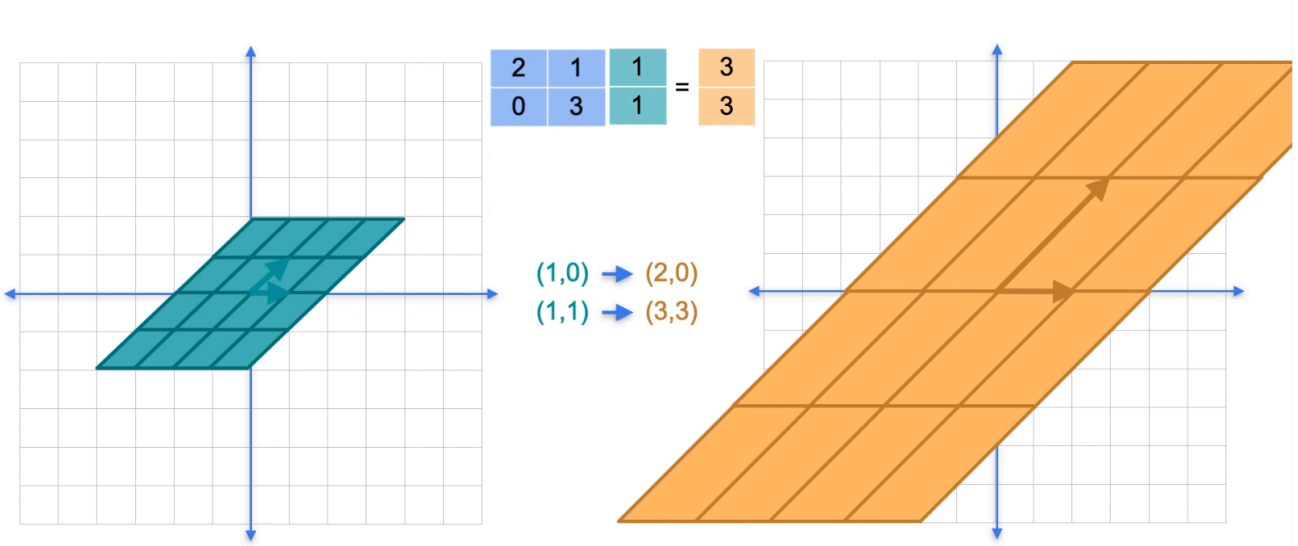
Span

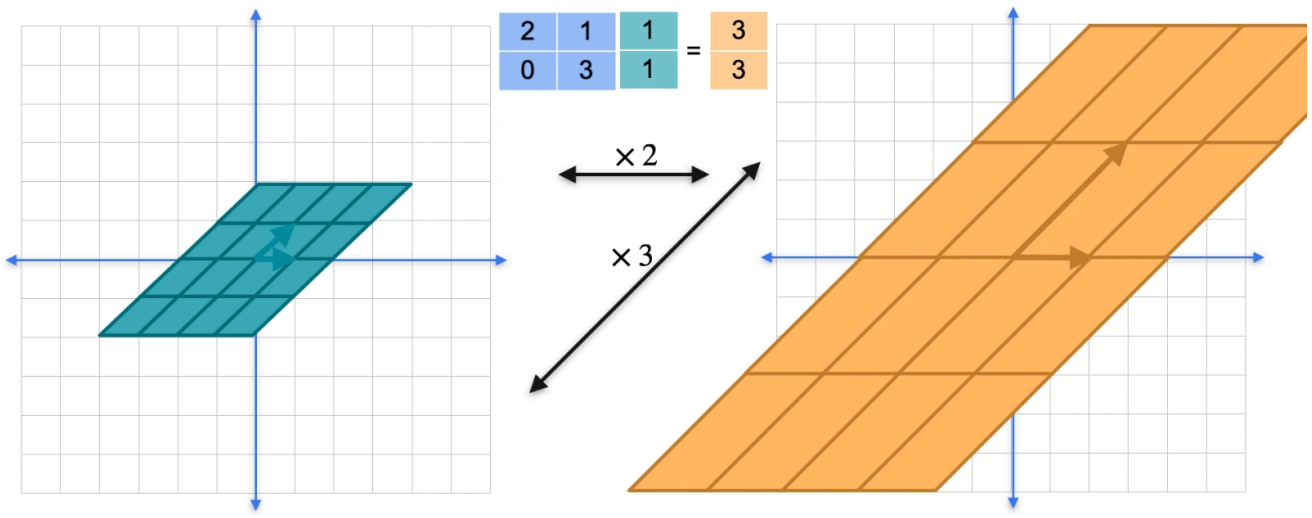
Span

Eigenbasis



The size of two parallelograms is parallel to the corresponding one on the other basis.
and the rest of the plan also follows





We stretch it by 2 times in the horizontal direction and 3 times in the diagonal direction

It only consists of 2 stretching, that is what is called an Eigenbasis

- The two vectors in the basis are going to be called the eigenvectors.

- The stretching factors 2 and 3 are going to be called eigenvalues.

$$Av = \lambda v$$

  The v is the matrix A's Eigenvector
  The λ is the matrix A's Eigenvalue
- Matrix Multiplication (left)= we have to do more work
- Scaler Multiplication (right)= we do less work, take only two multiplications
  All in all, with this equation, you can turn a large multiplication into a smaller one
- Eigenvalue = How much stretch
- Eigenvector = direction of the stretch
- Eigenbasis = the set of a matrix's eigenvectors, which can be arranged as a matrix with one eigenvector in each column

Calculating Eigenvalues and Eigenvectors

Finding the eigenvalues

If λ is an eigenvalue:

$$\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

For infinitely many (x,y)

$$\begin{bmatrix} 2-\lambda & 1 \\ 0 & 3-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Has infinitely many solutions

$$\det \begin{bmatrix} 2-\lambda & 1 \\ 0 & 3-\lambda \end{bmatrix} = 0$$

Characteristic polynomial $(2-\lambda)(3-\lambda) - 1 \cdot 0 = 0$

$$\lambda = 2$$
$$\lambda = 3$$

Finding the eigenvector

Eigenvalues: $\lambda = 2$, $\lambda = 3$

Solve the equations

$$\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 2 \begin{bmatrix} x \\ y \end{bmatrix} \qquad \begin{aligned} 2x + y &= 2x \\ 0x + 3y &= 2y \end{aligned} \qquad \begin{aligned} x &= 1 \\ y &= 0 \end{aligned} \qquad \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 3 \begin{bmatrix} x \\ y \end{bmatrix} \qquad \begin{aligned} 2x + y &= 3x \\ 0x + 3y &= 3y \end{aligned} \qquad \begin{aligned} x &= 1 \\ y &= 1 \end{aligned} \qquad \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Example: 3 Dims

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & -3 \\ -1 & -3 & 0 \end{bmatrix} \qquad \lambda I = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

Characteristic polynomial: $\det(A - \lambda I) = 0$

$$\det \begin{bmatrix} 2-\lambda & 1 & -1 \\ 1 & -\lambda & -3 \\ -1 & -3 & -\lambda \end{bmatrix} = 0$$

$$(2-\lambda)\lambda^2 \;\; +3 \;\; +3 \;\; -9(2-\lambda) \;\; +\lambda \;\; +\lambda \;\; = -\lambda^3 + 2\lambda^2 + 11\lambda - 12 = 0$$

$$-(\lambda+3)(\lambda-1)(\lambda-4) = 0$$

Eigenvalues: $-3, 1, 4$

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & -3 \\ -1 & -3 & 0 \end{bmatrix} \qquad \textbf{Eigenvalues: } -3, 1, \boxed{4}$$

$$Av = \lambda v$$

$$\underbrace{\begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & -3 \\ -1 & -3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\begin{bmatrix} 2x_1 + x_2 - x_3 \\ x_1 - 3x_3 \\ -x_1 - 3x_2 \end{bmatrix}} = 4 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4x_1 \\ 4x_2 \\ 4x_3 \end{bmatrix}$$

$$-x_1 - 3x_2 = 4x_3$$

$$\begin{array}{lrcl} R_1 & -2x_1 + x_2 - x_3 & = & 0 \\ R_2 & x_1 - 4x_2 - 3x_3 & = & 0 \\ R_3 & -x_1 - 3x_2 - 4x_3 & = & 0 \end{array}$$

$$R_2 + R_3 \qquad\qquad 3R_1 + R_3$$

$$-7x_2 - 7x_3 = 0 \qquad\qquad -7x_1 - 7x_3 = 0$$

$$x_2 = -x_3 \qquad\qquad x_1 = -x_3$$

$$\begin{array}{c} x_1 = k \\ x_2 = k \\ x_3 = -k \end{array} \qquad \begin{array}{c} x_1 = 1 \\ x_2 = 1 \\ x_3 = -1 \end{array} \qquad \begin{array}{c} x_1 = 2 \\ x_2 = 2 \\ x_3 = -2 \end{array}$$

infinite solutions of this form — this works! — so does this!

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & -3 \\ -1 & -3 & 0 \end{bmatrix}$$

Eigenvalues	$\lambda_1 = 4$	$\lambda_2 = 1$	$\lambda_3 = -3$
Eigenvectors	$\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$

Note:

- The determinant only works on a square matrix
- The eigenvalues can be repeated
- When the vectors point in different directions, they will have different eigenvectors

$$A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 4 & -0.5 \\ 0 & 0 & 2 \end{bmatrix}$$

Eigenvalues	$\lambda_1 = 4$	$\lambda_2 = 2$	$\lambda_3 = 2$
Eigenvectors	$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$

- We can choose any numbers for the vector if the root of the equation can take any numbers

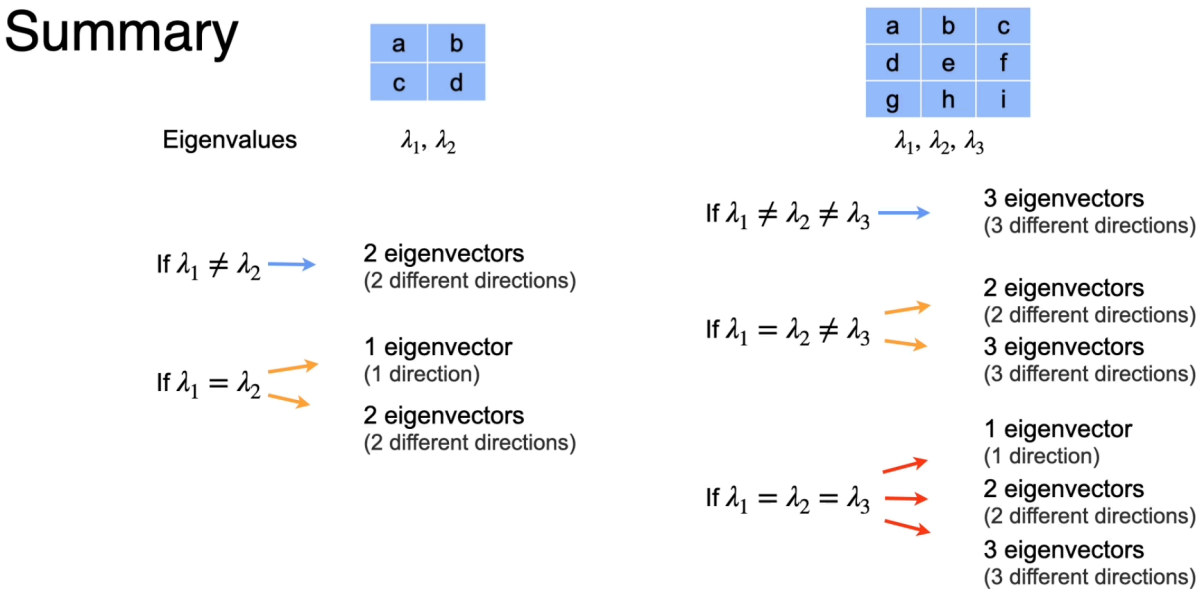
$$A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 4 & -0.5 \\ 4 & 0 & 2 \end{bmatrix} \quad \text{Eigenvalue: } 4$$

$$Av = 4v$$

$$\underbrace{\begin{bmatrix} 2 & 0 & 0 \\ -1 & 4 & -0.5 \\ 4 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\begin{bmatrix} 2x_1 \\ -x_1 + 4x_2 - 0.5x_3 \\ 4x_1 + 2x_3 \end{bmatrix}} = 4 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4x_1 \\ 4x_2 \\ 4x_3 \end{bmatrix}$$

$$\begin{aligned} 2x_1 &= 4x_1 \\ -x_1 + 4x_2 - 0.5x_3 &= 4x_2 \\ 4x_1 + 2x_3 &= 4x_3 \end{aligned}$$

$$\begin{aligned} -2x_1 &= 0 \\ -x_1 - 0.5x_3 &= 0 \\ 4x_1 - 2x_3 &= 0 \end{aligned}$$

$x_1 = 0 \qquad x_3 = 0 \qquad x_2 =$ any number

Summary